

SAMEDAM, Switzerland

WMO: 067920

Lat: 46.5261N

Lon: 9.8794E

Elev: 5610

StdP: 11.95

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -5.0	(d) -15.2	(e) 3.0	(f) -5.0	(g) -11.2	(h) 3.7	(i) -3.8	(j) 20.1	(k) 31.8	(l) 17.3	(m) 28.3	(n) 2.3	(o) 200	(p) 0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.6	(c) 73.6	(d) 56.6	(e) 70.8	(f) 55.3	(g) 68.1	(h) 54.2	(i) 58.4	(j) 69.7	(k) 56.9	(l) 67.6	(m) 55.5	(n) 65.5	(o) 11.4	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.0	76.6	62.3	52.5	72.4	60.6	51.1	68.7	59.1	28.0	69.7	26.8	67.7	25.8	65.6	64.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a) 18.6	(b) 16.6	(c) 14.6	(e) -16.8	(f) 77.4	(g) 5.3	(h) 2.8	(i) -20.6	(j) 79.4	(k) -23.7	(l) 81.0	(m) -26.7	(n) 82.5	(o) -30.6	(p) 84.5
			(e) -17.1	(f) 61.1	(g) 5.2	(h) 1.5	(i) -20.9	(j) 62.2	(k) -24.0	(l) 63.1	(m) -26.9	(n) 64.0	(o) -30.7	(p) 65.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	17.1	19.5	27.8	36.0	44.5	51.1	54.3	53.7	46.8	40.1	29.0	19.8
	DBStd	14.87	8.78	8.45	7.66	6.01	5.09	5.62	4.81	4.68	5.30	6.23	8.58	9.13
	HDD50	5257	1020	854	689	421	182	55	18	20	123	309	629	937
	HDD65	10319	1485	1274	1154	871	636	417	332	350	547	772	1079	1402
	CDD50	413	0	0	0	0	11	88	151	136	26	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	61	0	0	0	0	0	14	22	25	0	0	0	0
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0
	WSAvg	4.9	3.5	4.2	5.0	5.6	6.1	5.9	5.9	5.7	5.3	4.9	3.8	3.5
	PrecAvg	38.5	2.0	1.8	1.9	2.3	3.5	4.6	4.7	4.9	3.6	3.5	3.7	2.1
(15) Precipitation	PrecMax	52.5	4.7	5.9	4.9	5.4	7.8	8.8	8.9	7.6	8.4	9.1	14.4	5.9
	PrecMin	29.1	0.2	0.5	0.2	0.2	1.2	2.0	2.6	0.8	0.9	0.0	0.3	0.0
	PrecStd	6.7	1.3	1.4	1.1	1.2	1.5	1.9	1.4	1.5	2.0	2.0	3.3	1.3
	PrecStd	6.7	1.3	1.4	1.1	1.2	1.5	1.9	1.4	1.5	2.0	2.0	3.3	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	42.9	45.3	51.2	60.8	68.6	75.9	76.5	77.3	69.4	63.5	54.8	43.4	34.1
	0.4% MCWB	34.5	35.2	38.6	44.4	51.9	56.7	58.0	57.1	53.1	48.1	41.5	34.1	34.1
	2% DB	37.0	40.1	47.5	56.3	64.0	72.2	73.7	72.8	65.8	59.1	48.1	38.4	33.6
	2% MCWB	32.1	32.2	36.5	42.8	49.4	55.7	56.9	56.4	51.6	46.1	39.0	33.6	33.6
	5% DB	34.0	36.5	44.0	52.0	60.6	68.9	70.7	69.7	62.5	55.6	44.1	35.8	32.6
	5% MCWB	30.6	30.6	34.9	40.5	47.7	54.2	55.3	55.3	50.9	45.0	37.9	32.6	32.6
	10% DB	31.5	33.4	40.5	48.0	57.2	65.3	67.8	66.5	59.1	52.1	40.6	33.6	31.0
	10% MCWB	28.7	29.2	33.5	38.7	45.9	52.8	54.3	54.5	49.2	44.1	36.5	31.0	31.0
	0.4% WB	35.9	35.9	40.0	46.1	53.5	59.6	60.7	60.3	55.9	50.6	44.1	36.9	36.9
	0.4% MCDB	40.6	42.8	49.2	56.8	65.4	71.7	71.8	71.8	65.2	58.7	49.8	39.8	39.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.1	33.5	37.6	43.9	50.8	57.2	58.6	58.2	53.8	48.4	41.1	34.7	34.7
	2% MCDB	36.0	38.6	44.3	53.9	61.9	68.8	70.0	68.8	62.3	55.2	46.3	37.1	37.1
	5% WB	31.1	31.4	35.8	41.5	48.8	55.4	56.8	56.7	52.1	46.7	38.7	33.0	33.0
	5% MCDB	33.7	35.5	42.1	50.2	58.2	66.4	67.5	66.8	59.8	52.6	42.6	35.5	35.5
	10% WB	28.8	29.5	34.1	39.4	47.0	53.7	55.4	55.2	50.3	45.1	36.8	31.0	31.0
	10% MCDB	31.4	33.4	39.7	46.7	55.1	63.6	65.4	64.3	57.3	50.5	40.1	33.7	33.7
(35) Mean Daily Temperature Range	MDBR	22.7	25.3	21.9	19.6	20.5	22.7	23.6	22.4	21.7	21.7	18.7	20.0	20.0
	5% DB	21.8	28.3	26.5	27.4	28.3	30.0	29.7	29.6	28.6	28.9	23.9	19.9	19.9
	MCWBR	16.9	20.8	16.6	15.3	15.5	15.5	14.7	15.0	15.8	16.8	15.5	14.8	14.8
	5% WB	18.2	24.4	22.2	24.4	25.3	27.2	26.5	26.0	24.2	22.1	19.2	16.5	16.5
	MCWBR	14.5	18.1	14.2	14.0	14.1	14.5	13.7	13.9	14.0	13.6	13.0	12.7	12.7
	MCWBR	14.5	18.1	14.2	14.0	14.1	14.5	13.7	13.9	14.0	13.6	13.0	12.7	12.7
(40) Clear-Sky Solar Irradiance	taub	0.229	0.252	0.294	0.342	0.356	0.365	0.359	0.352	0.329	0.300	0.260	0.228	0.228
	taud	2.337	2.272	2.223	2.212	2.302	2.322	2.338	2.378	2.427	2.483	2.459	2.370	2.370
	Ebn at Noon	284	296	296	289	288	285	285	283	281	275	269	272	272
	Edh at Noon	23	30	37	42	40	39	38	35	30	25	20	20	20
(44) All-Sky Solar Radiation	RadAvg	513	820	1208	1425	1531	1782	1789	1555	1210	807	498	426	426
	RadStd	60	94	164	219	158	149	146	139	119	99	77	54	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.03	+1.04	+0.97	N/A	-259	+90	N/A
(47) Regional (107 neighbors)	+0.59	N/A	N/A	+1.19	+0.74	+0.63	N/A	N/A	+108	+27

Nomenclature: See separate page